RECURSION

Recursion Example

- A **recursive algorithm** is one that invokes (makes reference to) itself repeatedly until a certain condition (also known as termination condition) matches

```
void recursion() {
   recursion(); /* function calls itself */
}

int main() {
   recursion();
}
```

Recursion VS Iteration

- An Iterative algorithm **will be faster** than the Recursive algorithm because of overheads like calling functions and registering stacks repeatedly.
- Recursive algorithm uses a **branching structure**, while iterative algorithm uses a **looping construct**.
- **Recursive algorithms** are not efficient as they take more space and time.
- Recursive algorithms are mostly used **to solve complicated problems** when their application is easy and effective.
- Some problems are **inherently recursive**, like tree traversal, quick sort, merge sort, etc.

Factorial – A case study

- The factorial of a positive number is the product of the integral values from 1 to the number:

$$n! = n*(n-1)*(n-2)……….3*2*1$$

$$4! = 4.3.2.1 = 24$$

- **Iterative Factorial Algorithm definition:**

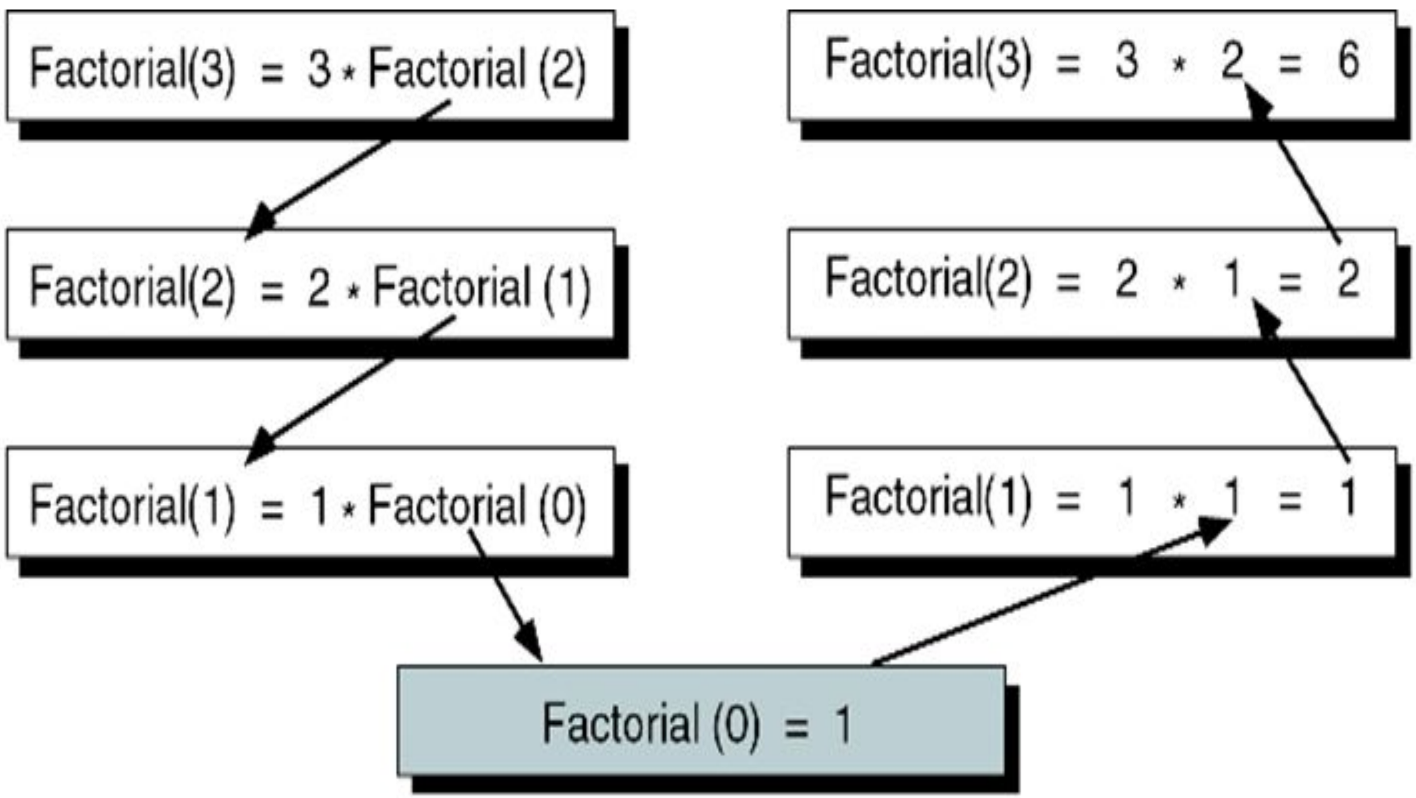
$$\text{Factorial}(n) = \begin{bmatrix} 1 & \text{if } n = 0 \\ n \times (n-1) \times (n-2) \times \ldots \times 3 \times 2 \times 1 & \text{if } n > 0 \end{bmatrix}$$

- **Recursive Factorial Algorithm definition:**

$$\text{Factorial}(n) = \begin{bmatrix} 1 & \text{if } n = 0 \\ n \times (\text{Factorial}(n-1)) & \text{if } n > 0 \end{bmatrix}$$

Factorial (3): Decomposition and solution

- The recursive solution for a problem involves a two-way journey:
  - First we decompose the problem from the top to the bottom
  - Then we solve the problem from the bottom to the top



ALGORITHM 2-1 Iterative Factorial Algorithm

```
Algorithm iterativeFactorial (n)
Calculates the factorial of a number using a loop.
   Pre  n  is the number to be raised factorially
   Post n! is returned
1 set i to 1
2 set factN to 1
3 loop (i <= n)
   1  set factN to factN * i
   2  increment i
4 end loop
5 return factN
end iterativeFactorial
```

ALGORITHM 2-2 Recursive Factorial

```
Algorithm recursiveFactorial (n)
Calculates factorial of a number using recursion.
   Pre    n  is the number being raised factorially
   Post   n! is returned
1 if (n equals 0)
   1  return 1
2 else
   1  return (n * recursiveFactorial (n - 1))
3 end if
end recursiveFactorial
```

Example-Problem GCD

- Determine the greatest common divisor (GCD) for two numbers.
- Euclidean algorithm: GCD (a,b) can be recursively found from the formula

$$GCD(a,b) = \begin{cases} a & \text{if } b=0 \\ b & \text{if } a=0 \\ GCD(b, a \bmod b) & \text{otherwise} \end{cases}$$

ALGORITHM 2-4 Euclidean Algorithm for Greatest Common Divisor

```
Algorithm gcd (a, b)
Calculates greatest common divisor using the Euclidean algo-
rithm.
   Pre  a and b are positive integers greater than 0
   Post greatest common divisor returned
1 if (b equals 0)
   1  return a
2 end if
3 if (a equals 0)
   2  return b
4 end if
5 return gcd (b, a mod b)
end gcd
```

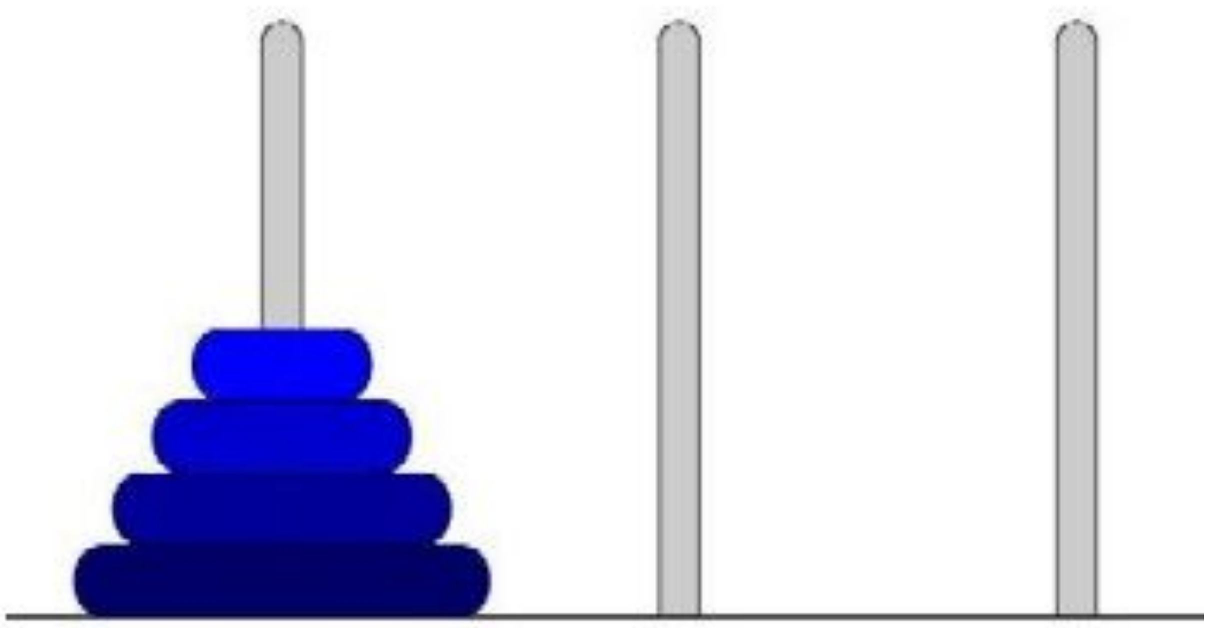
Example-Problem Fibonacci series

- Generation of the Fibonacci numbers series.
- Each next number is equal to the sum of the previous two numbers.
- A classical Fibonacci series is 0, 1, 1, 2, 3, 5, 8, 13, …
- The series of n numbers can be generated using a recursive formula

$$Fibonacci(n) = \begin{cases} 0 & \text{if } n=0 \\ 1 & \text{if } n=1 \\ Fibonacci(n-1) + Fibonacci(n-2) & \text{otherwise} \end{cases}$$

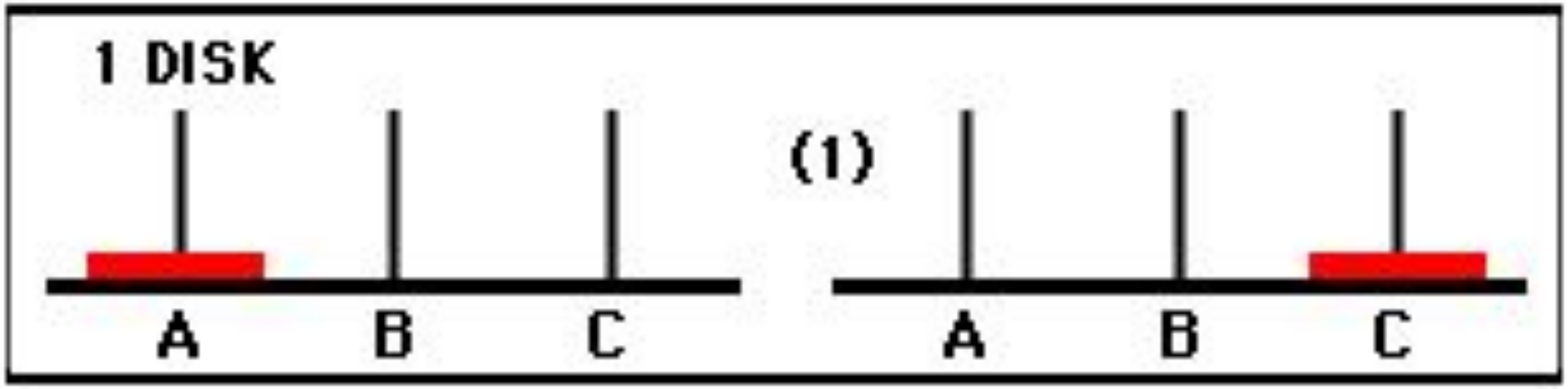
TOWERS OF HANOI

- Disks of different sizes (call the number of disks "n") are placed on the left-hand post,
- Arranged by size with the smallest on top.
- Your job is to transfer all the disks to the right-hand post in the fewest possible moves without ever placing a larger disk on a smaller one.



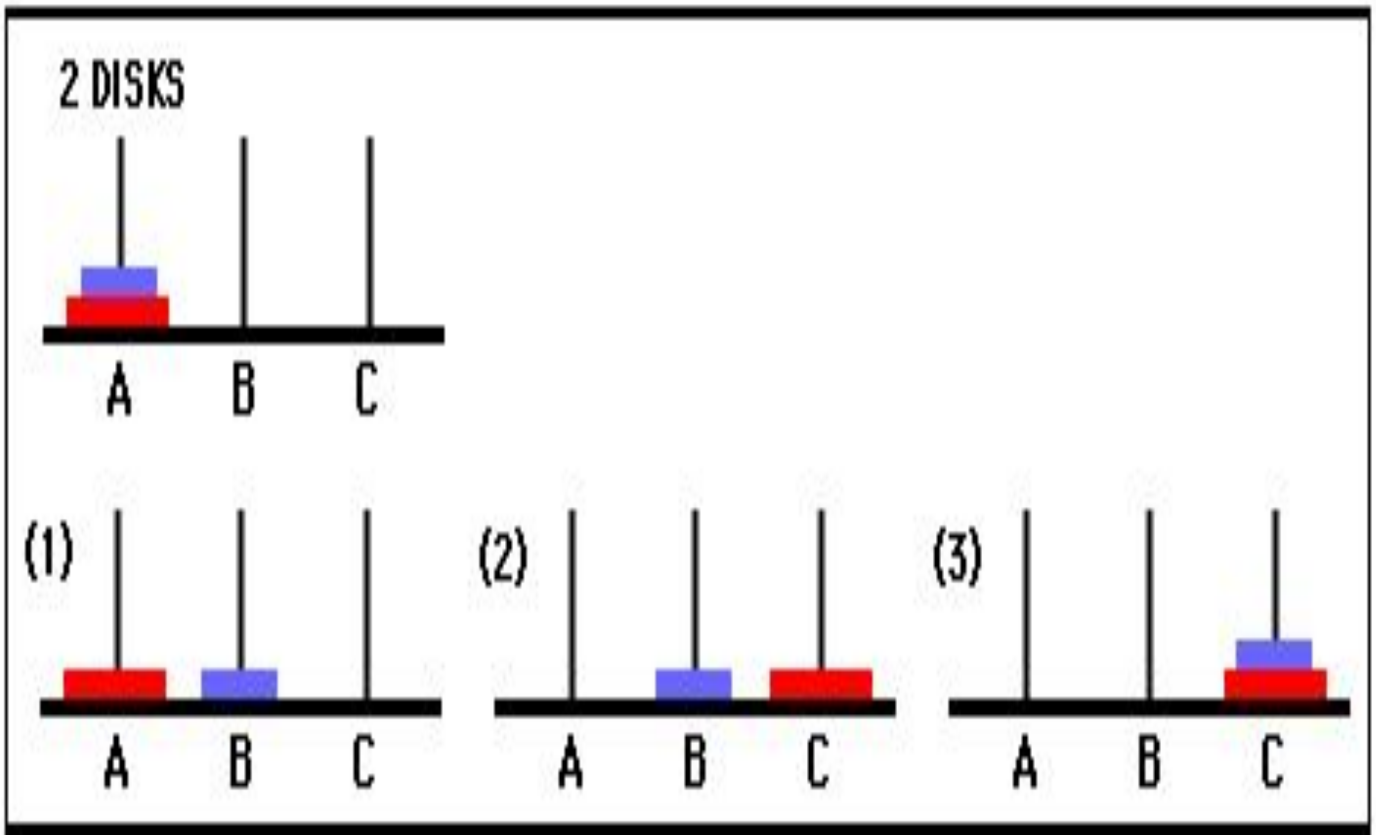
TOWERS OF HANOI

- **How many moves will it take to transfer n disks from the left post to the right post?**
- Let's look for a pattern in the number of steps it takes to move just one, two, or three disks. We'll number the disks starting with disk 1 on the bottom.**1 disk: 1 move**
- Move 1: move disk 1 to post C



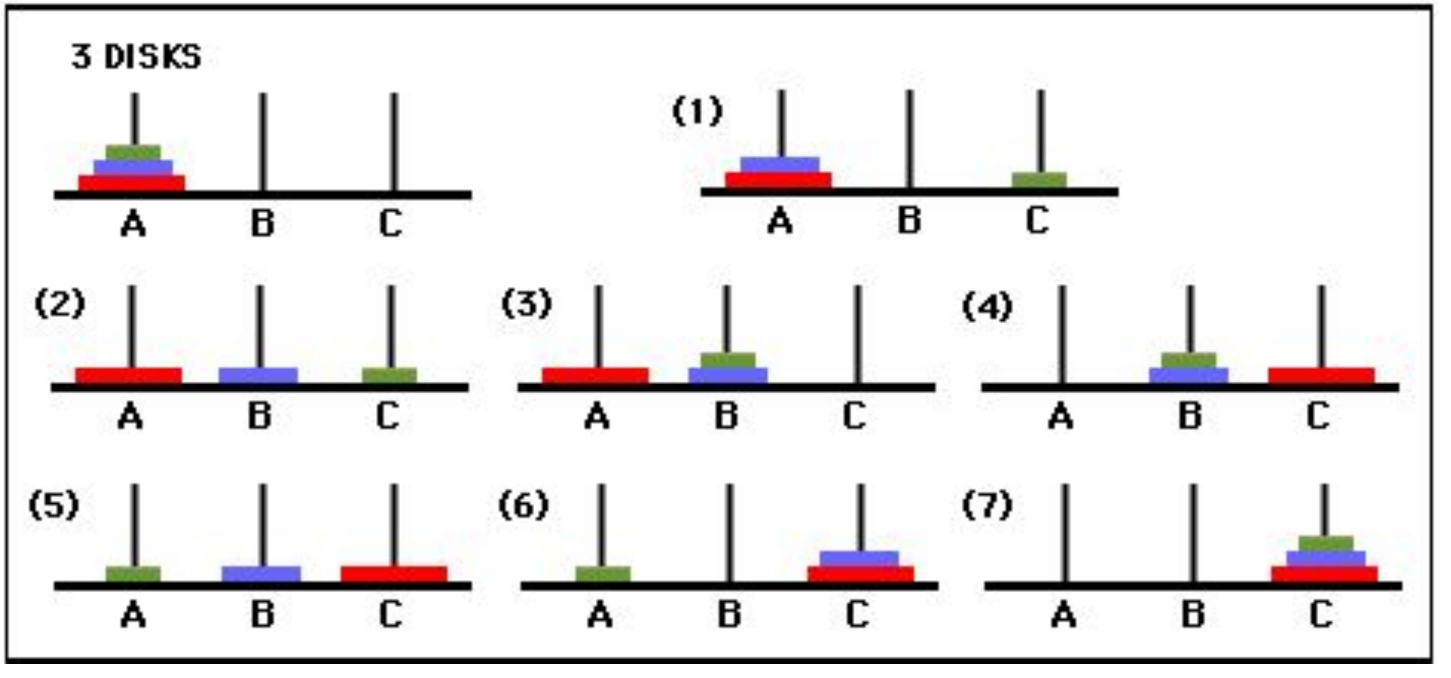
2 disks: 3 moves

Move 1: move disk 2 to post B
Move 2: move disk 1 to post C
Move 3: move disk 2 to post C



3 disks: 7 moves

Move 1: move disk 3 to post C
Move 2: move disk 2 to post B
Move 3: move disk 3 to post B
Move 4: move disk 1 to post C
Move 5: move disk 3 to post A
Move 6: move disk 2 to post C
Move 7: move disk 3 to post C



Example: Recursive solution to the Towers of Hanoi problem for N=3.

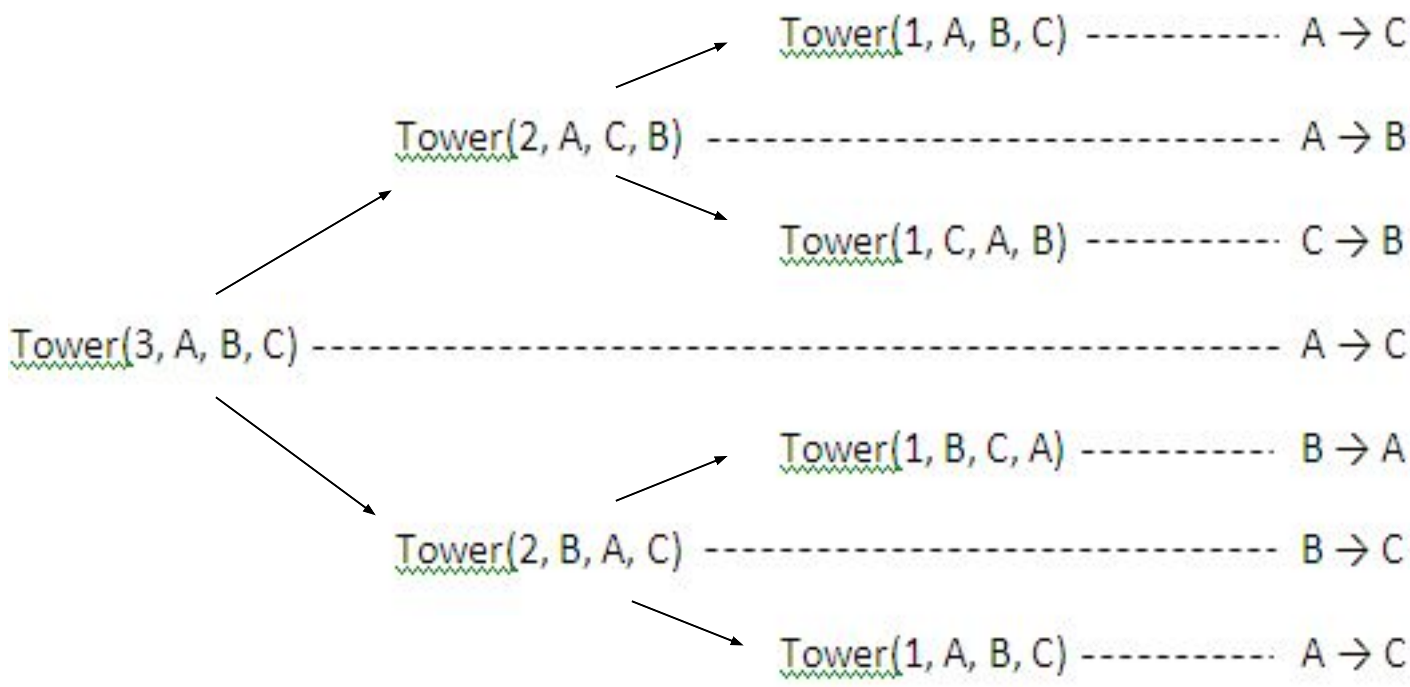


Figure: Recursive solution to the Towers of Hanoi problem for N=3.

<u>Assignment:</u> Write a recursive solution to the Towers of Hanoi for N=5.

TOWER(N, BEG, AUX, END)

This procedure gives a recursive solution to the Towers of Hanoi problem for N disks.

1. If N=1, then:
   (a) Write : BEG   END
   (b) Return.
   [End of If structure]
2. [Move N-1 disks from peg BEG to peg AUX]
   call TOWER(N-1, BEG, END,AUX)
3. Write : BEG   END
4. [Move N-1 disks from peg AUX to peg END]
   call TOWER(N-1, AUX, BEG, END)
5. Return.

Review Questions

- Design a recursive procedure to solve the Towers of Hanoi problem with n disks. Then trace your procedure to find a way to move four disks from tower BEG to END using an auxiliary tower AUX.
- Write a recursive function to compute the sum of the digits of a given number.
- Implement a recursive function to compute the 16th Fibonacci number, Fib(16) given that Fib(11) = 89 and, Fib(12) = 144. Write an iterative function to do the same. Compare between their time and space complexity.
- Find gcd(1238,780) using the Euclidean algorithm. What is the base case in the recursive GCD algorithm? Why is it important? Modify the given algorithm to count how many times the recursive function is called to get to the base case for a given pair of values.